

SSC MATH PRACTICE SET

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- Q1. A train 280 m long is running with a speed of 63 km/hr. In what time will it pass a man walking at 9 km/hr in the same direction?
- Q2. A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. The rate of interest per annum is:
- Q3. If $a:b = 2:3$ and $b:c = 4:5$, find $a:b:c$.
- Q4. A can do a piece of work in 10 days and B can do it in 15 days. How many days will both take together to complete the work?
- Q5. What is the value of $(a^3 - b^3) / (a - b)$ if $a = 101$ and $b = 100$?
- Q6. The average of 5 consecutive numbers is 20. Find the largest number.
- Q7. If 10% of x is the same as 20% of y , then $x:y$ is equal to:
- Q8. A shopkeeper marks his goods 20% above the cost price and allows a discount of 10%. His gain percent is:
- Q9. A and B invest in a business in the ratio 3:2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:
- Q10. If $\sin \theta = 3/5$, then find the value of $\tan \theta$.
- Q11. If $x + 1/x = 2$, find the value of $x^2 + 1/x^2$.
- Q12. The area of a circle is 616 cm^2 . Find its circumference.
- Q13. The ratio of the radii of two circles is 3:4. Find the ratio of their areas.
- Q14. A man sells an article at a loss of 20%. If he sells it for Rs. 12 more, he would have gained 10%. Find the cost price of the article.
- $0.8x + 12 = 1.1x$
- Q15. What is the value of $(0.2 \times 0.2 \times 0.2 + 0.3 \times 0.3 \times 0.3) / (0.2 \times 0.2 - 0.2 \times 0.3 + 0.3 \times 0.3)$?
- Q16. If $x - y = 4$ and $xy = 21$, find the value of $x^3 - y^3$.
- Q17. A boat can travel 20 km downstream in 2 hours and 4 km upstream in 2 hours. What is the speed of the boat in still water?
- Q18. Find the LCM of 12, 15 and 18.
- Q19. If the average of 7 numbers is 35 and the average of the first 3 numbers is 27, find the average of the last 4 numbers.
- Q20. What is the compound interest on Rs. 12000 for 2 years at 5% per annum?
- Q1. If a train 110 meters long passes a telegraph pole in 3 seconds, then the time taken by it to cross a railway platform 165 meters long is?
- Q2. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is?

- Q3. What is the value of $(0.04 * 0.04 * 0.04 - 0.03 * 0.03 * 0.03) / (0.04 * 0.04 + 0.04 * 0.03 + 0.03 * 0.03)$?**
- Q4. A and B can do a piece of work in 72 days; B and C can do it in 120 days; A and C can do it in 90 days. In what time can A alone do it?**
- Q5. The ratio between two numbers is 3:4 and their LCM is 60. The numbers are?**
- Q6. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?**
- Q7. If $x + 1/x = 2$, then $x^3 + 1/x^3 = ?$**
- Q8. If $2A = 3B = 4C$, then $A:B:C = ?$**
- Q9. A boat travels upstream from B to A and downstream from A to B in 3 hours. If the speed of the boat in still water is 9 km/hr and the speed of the stream is 3 km/hr, then the distance between A and B is?**
- Q10. The average of 5 numbers is 27. If one number is excluded, the average becomes 25. The excluded number is?**
- Q11. What is the area of an equilateral triangle with side $2\sqrt{3}$ cm?**
- Q12. The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is?**
- Q13. If $a = 23$ and $b = -29$, then find the value of $25a^2 + 40ab + 16b^2$.**
- Q14. If $x/y = 4/5$, then find the value of $(4x + 5y) / (5x - 2y)$.**
- Q15. A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely?**
- Q16. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is?**
- Q17. If the length of a rectangle is increased by 20% and the breadth is decreased by 20%, then the area?**
- Q18. The difference between compound interest and simple interest on a sum for 2 years at 8% per annum is Rs. 16. The sum is?**
- Q19. Two trains are moving in opposite directions at 60 kmph and 90 kmph respectively. Their lengths are 1.10 km and 0.9 km respectively. The time taken by the slower train to cross the faster train in seconds is?**
- Q20. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:**
- Q1. A train 280 m long is running with a speed of 63 km/hr. In what time will it pass a signal post?**
- Q2. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then find the value of x.**
- Q3. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:**
- Q4. A and B can do a piece of work in 12 days. B and C can do it in 15 days and C and A can**

do it in 20 days. How long would A take to do the work alone?

Q5. If $A : B = 2 : 3$ and $B : C = 4 : 5$, then $A : C = ?$

Q6. Two numbers are in the ratio 3:5. If 9 is subtracted from each, the new numbers are in the ratio 12:23. The smaller number is:

Q7. What is the value of $(0.2 \times 0.2 + 0.02 \times 0.02 - 0.4 \times 0.02) / 0.36$?

Q8. A shopkeeper marks his goods at 20% above the cost price and allows a discount of 10% on the marked price. His gain percent is:

Q9. The average of 5 consecutive numbers is n. If the next two numbers are also included, the average will:

Q10. What is the value of $\sin 30^\circ + \cos 60^\circ$?

Q11. If $x + \frac{1}{x} = 2$, then $x^3 + \frac{1}{x^3} = ?$

Q12. The area of a circle is 616 cm^2 . Find its circumference.

Q13. The HCF of two numbers is 11 and their LCM is 7700. If one of the numbers is 275, then the other is:

Q14. A man travels a certain distance at 6 km/hr and returns at 4 km/hr. If the total time taken is 10 hours, the distance is:

Q15. Simplify: $(a+b)^2 - (a-b)^2$

Q16. If 15% of x is equal to 20% of y, then $x:y = ?$

Q17. If the radius of a circle is increased by 50%, find the percentage increase in its area.

Q18. A person borrows Rs. 5000 for 2 years at 4% per annum simple interest. He immediately lends it to another person at $6 \frac{1}{4} \%$ per annum for 2 years. Find his gain in the transaction per year.

Q19. The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person?

Q20. A boat can travel 20 km downstream in 2 hours and 4 km upstream in 2 hours. What is the speed of the boat in still water?

Q1. A train 280 m long is running with a speed of 63 km/hr. In what time will it pass a tree?

Q2. If 15 oxen or 20 cows can eat the grass of a field in 80 days, how long would 6 oxen and 2 cows take to eat the same?

Q3. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price?

Q4. What is the value of $(a^3 + b^3)$ if $a + b = 10$ and $ab = 21$?

Q5. If $x + \frac{1}{x} = 2$, then find the value of $x^2 + \frac{1}{x^2}$.

Q6. What is the area of a circle whose circumference is 44 cm?

Q7. The average of 5 consecutive numbers is 27. Find the largest number.

Q8. A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. The rate of interest per annum is:

Q9. If $A:B = 2:3$ and $B:C = 4:5$, then find $A:C$.

- Q10.** The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:
- Q11.** Simplify: $(2^0 + 3^1 + 4^2) / (3^2 - 2^1)$
- Q12.** What is the HCF of 24, 36, and 40?
- Q13.** A can do a piece of work in 10 days, and B can do it in 15 days. How many days will they take to complete the work together?
- Q14.** If $\sin(A) = 3/5$, then find $\cos(A)$.
- Q15.** If the diameter of a sphere is 14 cm, find its volume.
- Q16.** What is the value of $(0.2)^3$?
- Q17.** A shopkeeper marks his goods 20% above the cost price and then allows a discount of 10%. Find his gain percent.
- Q18.** What is the area of an equilateral triangle with side 8 cm?
- Q19.** If $3x + 2y = 12$ and $xy = 6$, find the value of $9x^2 + 4y^2$.
- Q20.** The ratio of two numbers is 3:4 and their LCM is 60. Find the numbers.
- Q1.** A train 270 m long is running at 36 kmph. How much time will it take to cross a tree?
- Q2.** If $a = 2 + \sqrt{3}$, find the value of $a + (1/a)$.
- Q3.** What is the value of $((1/3) - (1/5)) / ((1/3) + (1/5))$?
- Q4.** If 15% of x is equal to 20% of y , then what is the ratio of $x:y$?
- Q5.** If the cost price of 15 articles is equal to the selling price of 12 articles, find the profit percent.
- Q6.** What is the simple interest on Rs. 5000 for 2 years at 5% per annum?
- Q7.** A sum of money amounts to Rs. 6690 after 3 years and to Rs. 10,035 after 6 years on compound interest. Find the sum.
- Q8.** If $x + y = 10$ and $x - y = 4$, then find the value of xy .
- Q9.** The average of 5 numbers is 20. If each number is multiplied by 2, then what will be the new average?
- Q10.** A and B can do a piece of work in 10 days. B and C can do it in 15 days, and C and A can do it in 20 days. How long will A alone take to do the work?
- Q11.** If $\sin \theta = 3/5$, find the value of $\cos \theta$.
- Q12.** Simplify: $(a^2 - b^2) / (a + b)$
- Q13.** If the diameter of a circle is 14 cm, find its area.
- Q14.** What is the value of $2^5 * 2^{-3}$?
- Q15.** The ratio of two numbers is 3:4 and their HCF is 4. Find their LCM.
- Q16.** A shopkeeper marks his goods 20% above the cost price and allows a discount of 10%. Find his gain percent.
- Q17.** Find the value of $(0.2)^2 + (0.02)^2$

Q18. A man travels a certain distance at 6 kmph and returns at 4 kmph. If the total time taken is 10 hours, find the distance.

Q19. What is the value of $\sqrt{144} + \sqrt{256}$?

Q20. If $a:b = 2:3$ and $b:c = 4:5$, then find $a:b:c$.